

RS1 Controller-Sound Combination Manual

Wolfgang Haring e.U.

Company Register Number: FN 312421 v (Regional Court Wr. Neustadt)

Heinrich Sauer-Gasse 21, A-2700 Wiener Neustadt

Phone: +49 30 6098490 431

Table of Contents

Welcome

Thank you for choosing one of our products. If you have any questions or suggestions, please do not hesitate to contact us via email at info@kraftwerk.shop or visit our website at www.kraftwerk.shop. There you will always find the latest user manuals, software updates, and instructional videos.

Enjoy your new product!

Safety Instructions

- Read this manual carefully, paying special attention to sections marked in red.
- Only perform wiring changes when the device is powered off.
- Ensure that unexpected reactions during startup or configuration cannot cause damage (e.g., keep drive wheels off the ground).
- Protect all electronic components from moisture and oil.
- Route data cables (antenna, bus cables) away from power cables (motor, battery, digital servos) to avoid interference.
- Replace any visibly damaged components or cables immediately.
- Operate components only within their specified technical limits.
- Disconnect the battery after use, especially important for LiPo batteries to prevent deep discharge.
- Avoid short circuits and reverse polarity connections.

Quick Overview

Software Version

Always download the latest version of the ControlPanel software and connect your module. All available updates, including new features and bug fixes, will be installed automatically.

Status LED Indicators

- **Normal operation:** LED blinks green briefly every two seconds.
- **Standby mode:** LED blinks yellow slowly; no control signals are accepted.
- **Fast red blinking:** Indicates undervoltage; warning lights activate and the motor shuts down.
- **Slow red or other colors:** Please contact support.

Standby Switch

The switch allows you to turn the model on/off or set it to Standby mode. In Standby, all functions continue running, but no control signals from the remote are accepted, effectively creating a demo mode.

In the OFF position, the controller and all connected servos and bus modules are switched off.

Warning: A small current still flows in OFF mode, which can lead to battery deep discharge if the model is stored in this state. Always disconnect the battery, at least at the end of the day.

Channel Learning

It is essential to learn the channels once. There are three methods:

- Without PC, as described in section 8.4 "Channel Learning".
- Using the learning assistant via ControlPanel: *System > Learn Channels*.
- Individually in the Live Data section.

Motor Start/Stop

Depending on your settings, you may need to start the motor manually. If the motor does not emit any sound, the drive motor will not turn.

Hissing Sounds

During operation, the brake is also controlled, accompanied by hissing sounds. This is intentional but can be adjusted or disabled in the ControlPanel.

Product Description

The RS1 combines an electronic speed controller, sound module, and lighting system in one compact unit. It connects to up to five channels of any receiver and supports SUMD, IBUS, allowing up to 32 channels within the system.

Robust Speed Controller

- Continuous current rating of 25A.
- 32kHz switching frequency suitable for bell-type motors.
- Includes clutch function and start assist.
- Support for gear shifting and cruise control will be enabled via future updates.

Exceptional Sound System

KraftwerkK sets new standards in programmability. While competitors allow only a few sound snippets for driving sounds, KraftwerkK supports an unlimited number of sound snippets, organized as follows:

1. Define the RPM range (e.g., 500 RPM after start, 2300 RPM at full throttle).
2. Assign sound snippets to multiple RPM bands with minimum, base, and maximum RPM values.
3. At minimum RPM, the sound fades in with RPM adaptation; at base RPM, it plays at full volume without RPM adjustment; and at maximum RPM, it fades out again with RPM adaptation.
4. Overlapping snippets create an ultra-realistic soundscape.

There are four sound tracks in total: two for acceleration and two for deceleration. The first track is the standard driving sound; the second can be mixed proportionally as load sound or switched. The third track is the deceleration sound, automatically mixed in when releasing the control stick, and the fourth is the engine brake sound.

If this sounds complex, rest assured that six high-quality sounds are preloaded on the module. Additionally, a sound library with premium sounds will be available at www.kraftwerk.shop, which can be purchased and downloaded via ControlPanel to your module.

Unlimited Expandability

KraftwerkK's leading role in RC bus systems shines with the RS1. It features six EasyBus outputs for unlimited expansion. Lighting boards, switch extensions, servo controls, and more can be controlled via EasyBus.

A special feature is the four small connectors on the top side, allowing additional modules to be stacked. The first of these is the BL8A, an 8A brushless controller designed for hydraulic pumps.

Factory Settings

Erste Ebene	Geschwindigkeit A -> B	Geschwindigkeit B -> A	Mittelstellung	Umkehren	Endausschlag A	Endausschlag B	Funktion	Funktionsgruppe Proportional	Proportionalfunktion	
Servoausgang 1	100 %	100 %	0 %	☐	100 %	100 %	Proportionalfunktion Direkt	PropFunktionen	Lenkung	
Erste Ebene	Geschwindigkeit A -> B	Geschwindigkeit B -> A	Mittelstellung	Umkehren	Endausschlag A	Endausschlag B	Funktion	Servostellung Gang 1	Servostellung Gang 2	Servostellung Gang 3
Servoausgang 2	100 %	100 %	0 %	☐	100 %	100 %	Schaltgetriebe	-100 %	0 %	100 %
Erste Ebene	Geschwindigkeit A -> B	Geschwindigkeit B -> A	Mittelstellung	Umkehren	Endausschlag A	Endausschlag B	Funktion			
Servoausgang 3	100 %	100 %	0 %	☐	100 %	100 %	Sattelplatte			
Erste Ebene	Geschwindigkeit A -> B	Geschwindigkeit B -> A	Mittelstellung	Umkehren	Endausschlag A	Endausschlag B	Funktion			
Servoausgang 4	100 %	100 %	0 %	☐	100 %	100 %	keine			
Erste Ebene	Geschwindigkeit A -> B	Geschwindigkeit B -> A	Mittelstellung	Umkehren	Endausschlag A	Endausschlag B	Funktion			
Servoausgang 4	100 %	100 %	0 %	☐	100 %	100 %	keine			
Erste Ebene	Einschalt- geschwindigkeit	Ausschalt- geschwindigkeit	Funktion	Standlicht						
Schaltausgang 1 (500mA)	100 %	100 %	Standlicht	20 %						
Erste Ebene	Einschalt- geschwindigkeit	Ausschalt- geschwindigkeit	Funktion	Abblendlicht	Xenoneffekt					
Schaltausgang 2 (500mA)	100 %	100 %	Abblendlicht	50 %	☐					
Erste Ebene	Einschalt- geschwindigkeit	Ausschalt- geschwindigkeit	Funktion	Fernlicht	Xenoneffekt					
Schaltausgang 3 (500mA)	100 %	100 %	Fernlicht	100 %	☒					
Erste Ebene	Einschalt- geschwindigkeit	Ausschalt- geschwindigkeit	Funktion							
Schaltausgang 4 (500mA)	100 %	100 %	Rundumlicht Ein/Aus							
Erste Ebene	Einschalt- geschwindigkeit	Ausschalt- geschwindigkeit	Funktion	Blinker						
Schaltausgang 5 (500mA)	100 %	100 %	Blinker links	100 %						
Erste Ebene	Einschalt- geschwindigkeit	Ausschalt- geschwindigkeit	Funktion	Blinker						
Schaltausgang 6 (500mA)	100 %	100 %	Blinker rechts	100 %						
Erste Ebene	Einschalt- geschwindigkeit	Ausschalt- geschwindigkeit	Funktion	Rücklicht	Bremslicht					
Schaltausgang 7 (500mA)	100 %	100 %	Rücklicht & Bremslicht	20 %	100 %					

image5.png

Connections

- 5 servo inputs; the Gas/Data input also accepts IBUS/SBUS/SUMD signals with automatic detection.
- 5 configurable servo outputs (factory settings detailed above).
- OFF/ON/Standby switch and jumper for configuring sound and control modes.
- 6 EasyBus outputs.
- I2C expansion connector.
- Micro USB port.
- SD card slot supporting up to 32GB FAT32 formatted cards.
- Speaker output (4Ω or higher impedance).
- 10A switching outputs.
- Status LED.
- 4 expansion pins for plug-in modules.

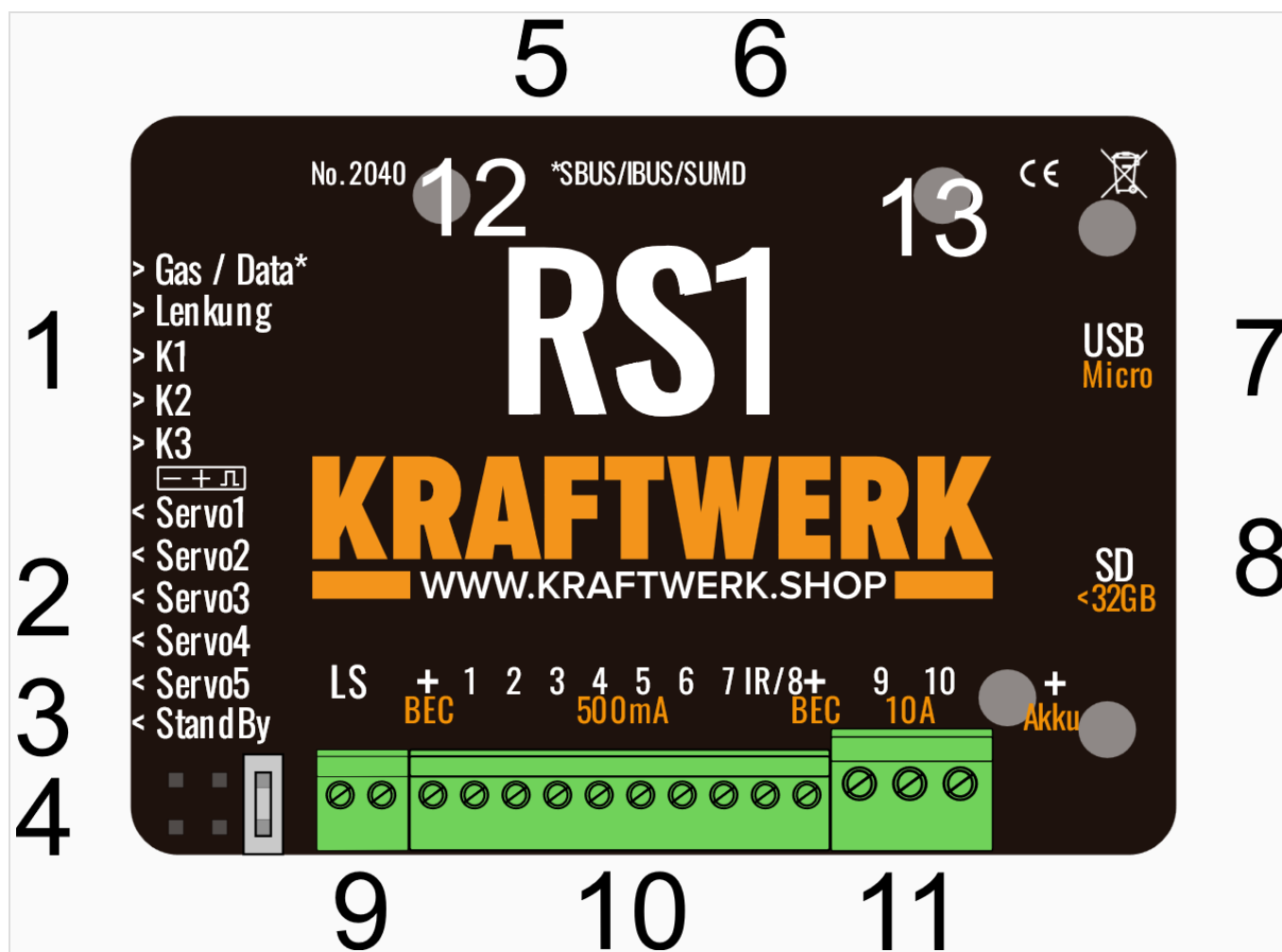


image6.png

Technical Specifications

Speed Controller

Parameter	Specification
Operating Voltage	6 - 20 V
BEC Output	6 V / 3 A
Switching Frequency	Adjustable up to 32 kHz
Clutch Function	Yes
Gear Shift Support	Yes (future update)
Inertia Simulation	Optional
Cruise Control	Optional
Protection Features	Battery undervoltage, motor short circuit, overtemperature
Servo Functions	Proportional control, two-position, three-position, saddle plate, trailer supports, motor for dump truck, ramp/platform motor, gear shift motor

Sound System

Parameter	Specification
Number of Sounds	Unlimited
Simultaneous Playback	Approx. 10

Playback Modes	Key, Loop, Once, Permanent, Random, Stop Points
Supported Formats	WAV, MP3
Sampling Rate	All (automatic conversion to 44,100 Hz)
Volume Control	Potentiometer, channel control, software
SD Card	Up to 32 GB FAT32; larger cards can be formatted to 32 GB



image8.png

Commissioning

The RS1 must be connected to five channels of the receiver. Using jumpers, the use of a control pad for lighting and sound can be configured.

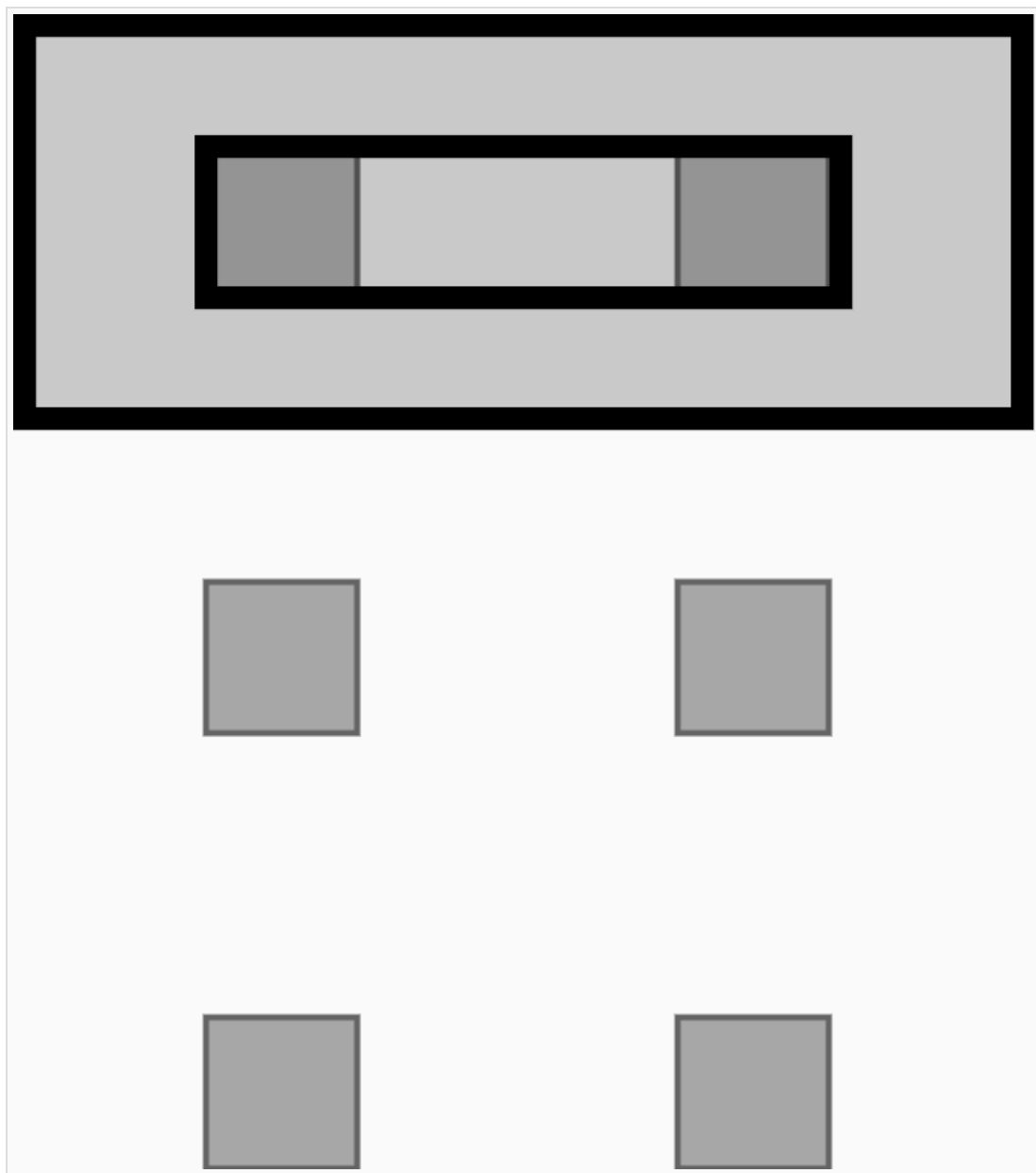


image9.png

When the right jumper is set, the control pad is connected to channel 3 (K3). The engine brake remains active as long as channel 2 (K2) is pushed up, provided the model sound supports this. If K2 is pushed down and then tapped right, the horn switches to horn and fanfare sounds, if supported by the model sound.

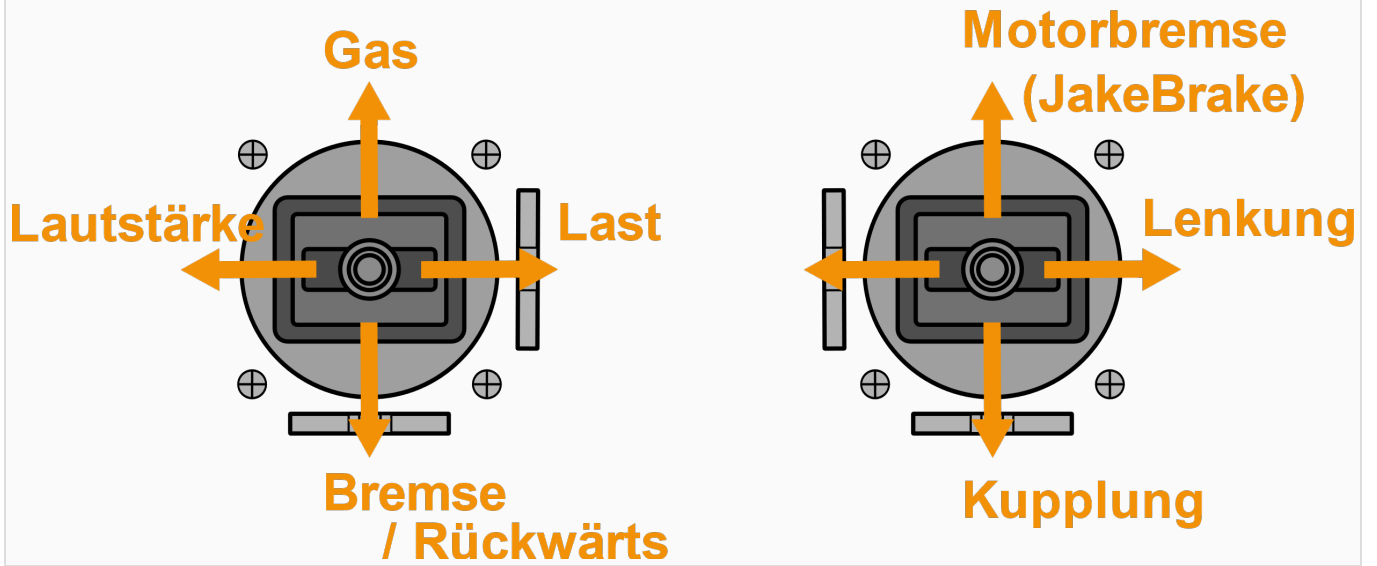
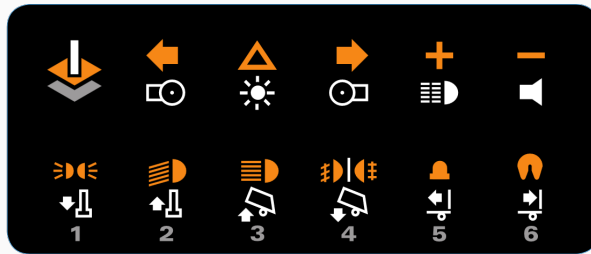


image11.png

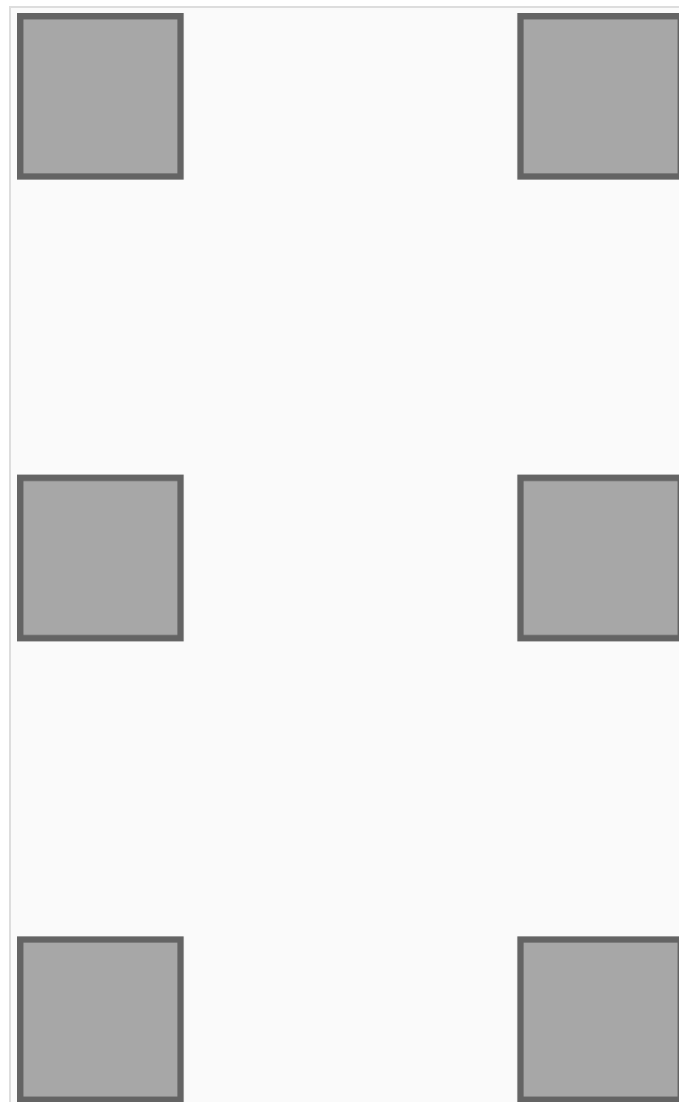


image13.png

If the upper jumper is not set, control is via buttons on K3. Without a control pad, channel assignments are denser. Throttle, steering, volume, and load operate as with the control pad. The engine brake functions as a memory toggle: a short tap activates it, another tap deactivates it. Holding the button up starts/stops the motor. Half taps activate the horn. Pushing K2 down and then tapping right switches the horn to horn and fanfare, if supported.

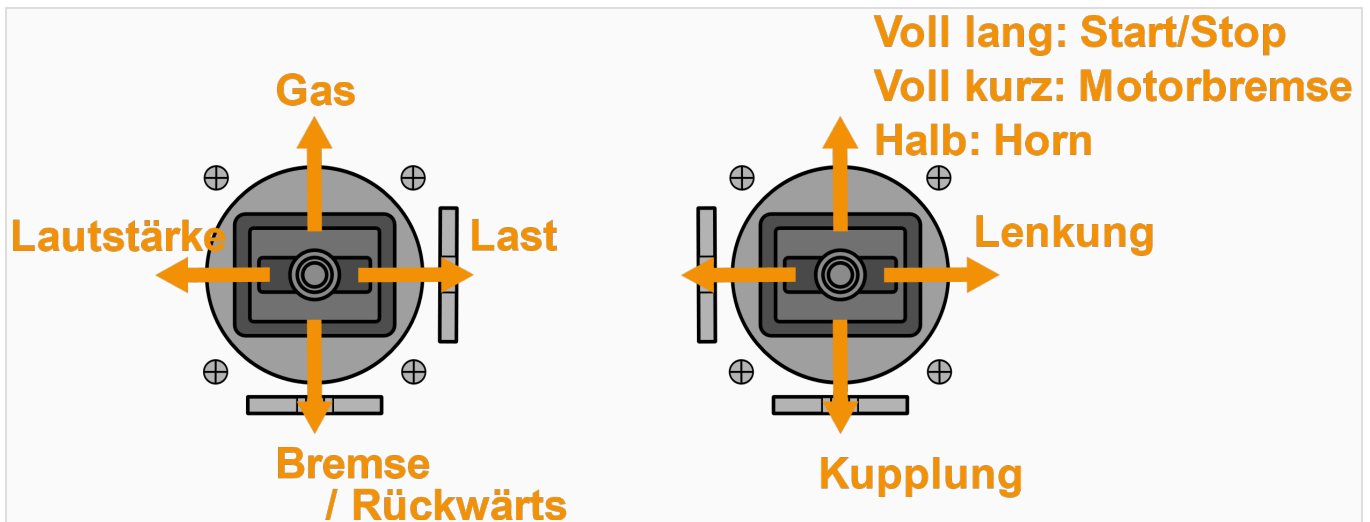


image15.png

Voll kurz: Blinker links
Voll lang: Warnblinker
Halb kurz: Sattelkupplung
Halb lang: PartyModus

image17.png

When using a button on K3, only full functions can be switched. With appropriate mixing on the remote control, half functions may be realized.

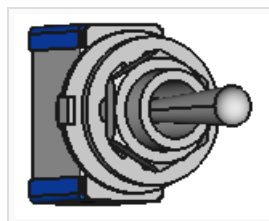


image19.png

Voll kurz: Blinker rechts
Voll lang: Licht+
Halb kurz: Rundumlicht
Halb lang: Licht aus

image20.png

Connecting to the Receiver



image8.png

Before connecting the RS1 to the receiver, verify your receiver's channel assignments. Use a servo to test each output for correct function and note the assignments.



image22.jpeg

Warning: It is recommended to elevate the model to prevent unintended movement during setup. Then connect all servo cables to the corresponding receiver outputs. If using an IBUS, SBUS, or SUMD receiver, the RS1 will automatically adjust. Channel assignments can be modified in Live Data if necessary.

Channel Learning Procedure



image8.png

Learning channels is crucial for proper operation of KraftwerK EasyBus systems. After any changes to your remote control settings (travel endpoints, directions, center points), channels must be relearned.

Servos and motors controlled by KraftwerK systems must be adjusted via ControlPanel, not the remote control menu.

1. Turn on the remote control and, if the drive motor is connected, elevate the model so wheels do not touch the ground.
2. Move throttle and steering sticks to their endpoints, then power on the model. The red LED lights up, indicating learning mode.
3. Return sticks to center position.
4. LED blinks to confirm each channel learning step:
 - 1 blink: Throttle full forward, then full brake.
 - 2 blinks: Steering full left, then full right.
 - 3 blinks: Auxiliary channel K1 full left, then full right.
 - 4 blinks: Auxiliary channel K2 full up, then full down.
 - 5 blinks: Auxiliary channel K3 full up, then full down.
5. When all sticks are centered again, the LED glows solid red, signaling the end of learning.
6. The system restarts automatically and signals readiness with double green LED flashes.

Model Sounds

Sound	Jumper Setting	Load Available	Engine Brake Available
-------	----------------	----------------	------------------------

Scania
770S

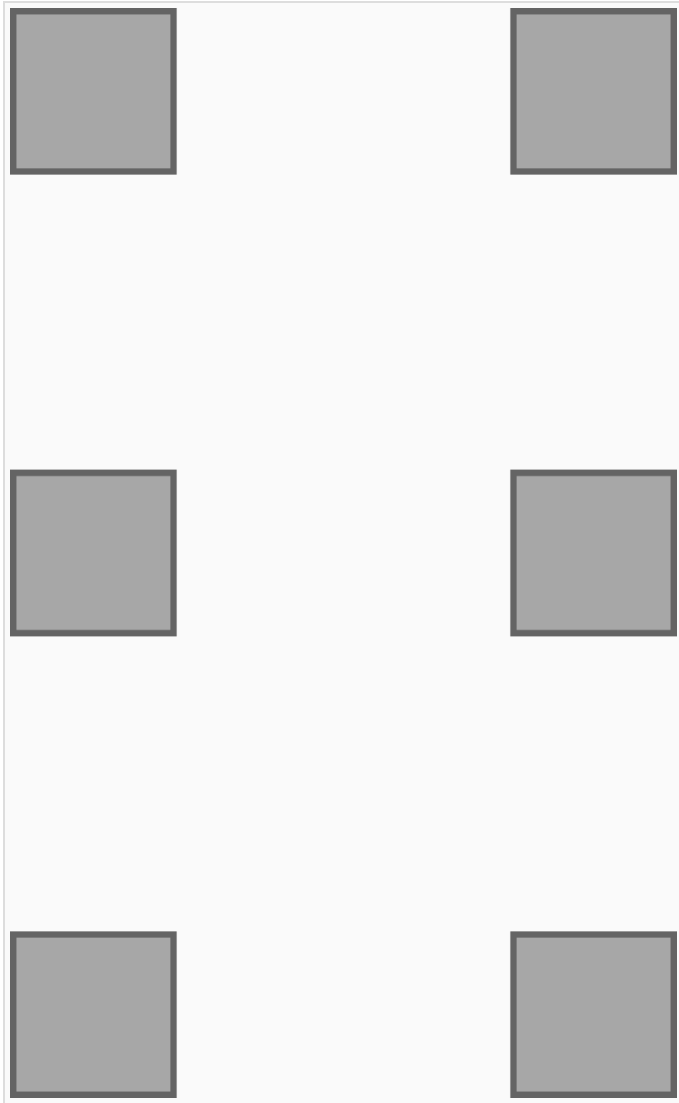


image13.png

Volvo FH16

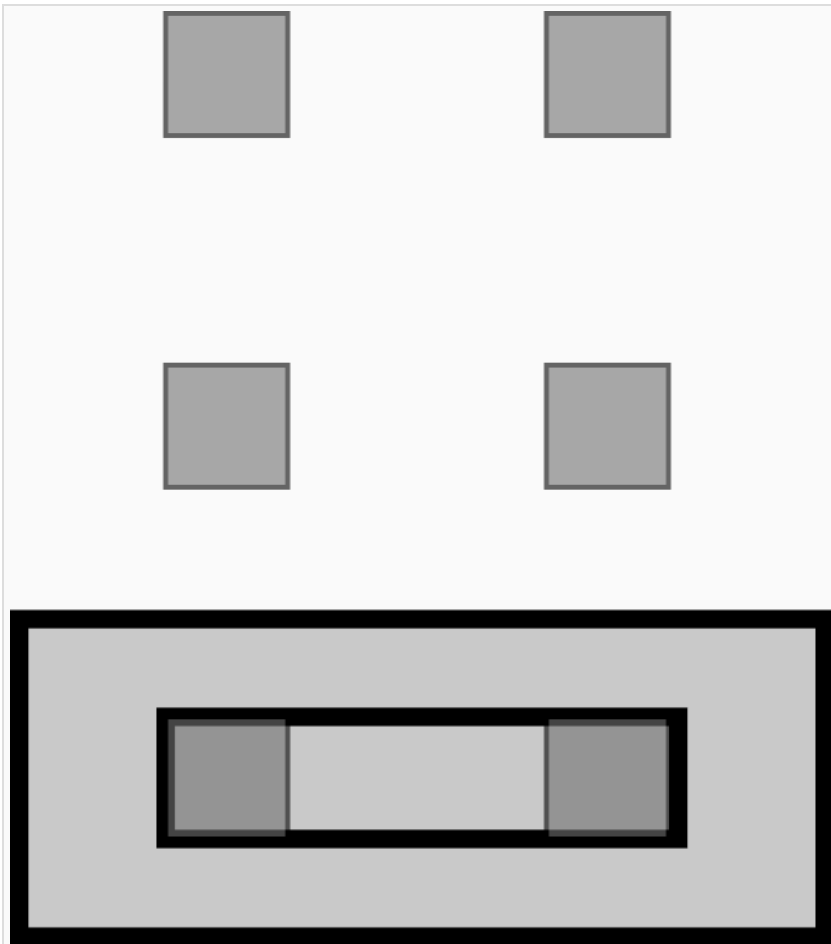


image24.png

Mercedes
Actros

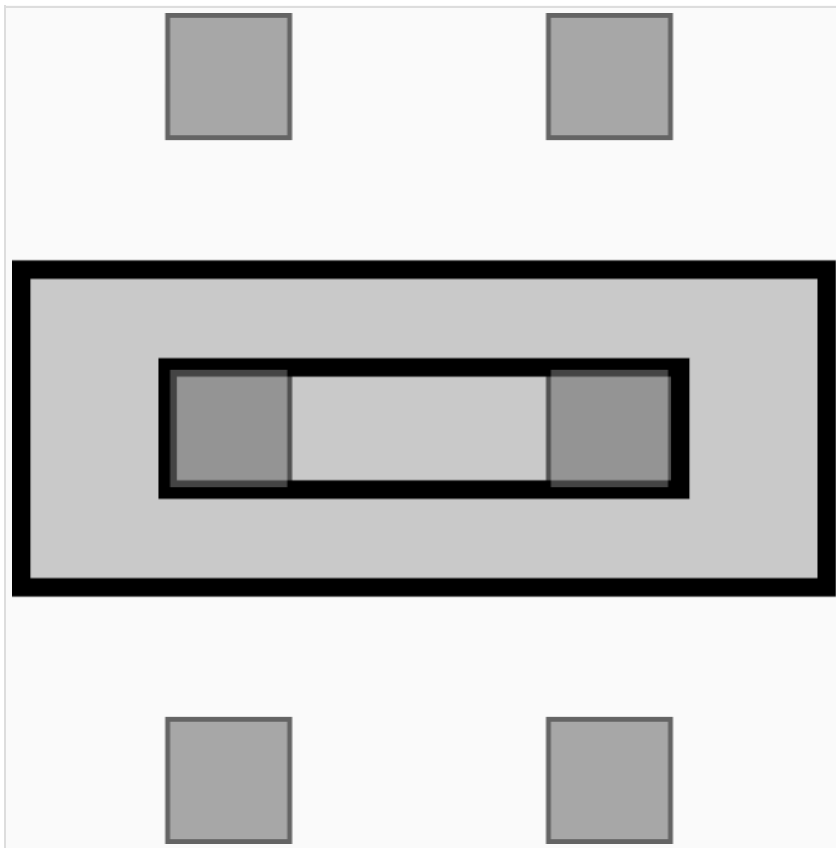


image26.png

Unimog

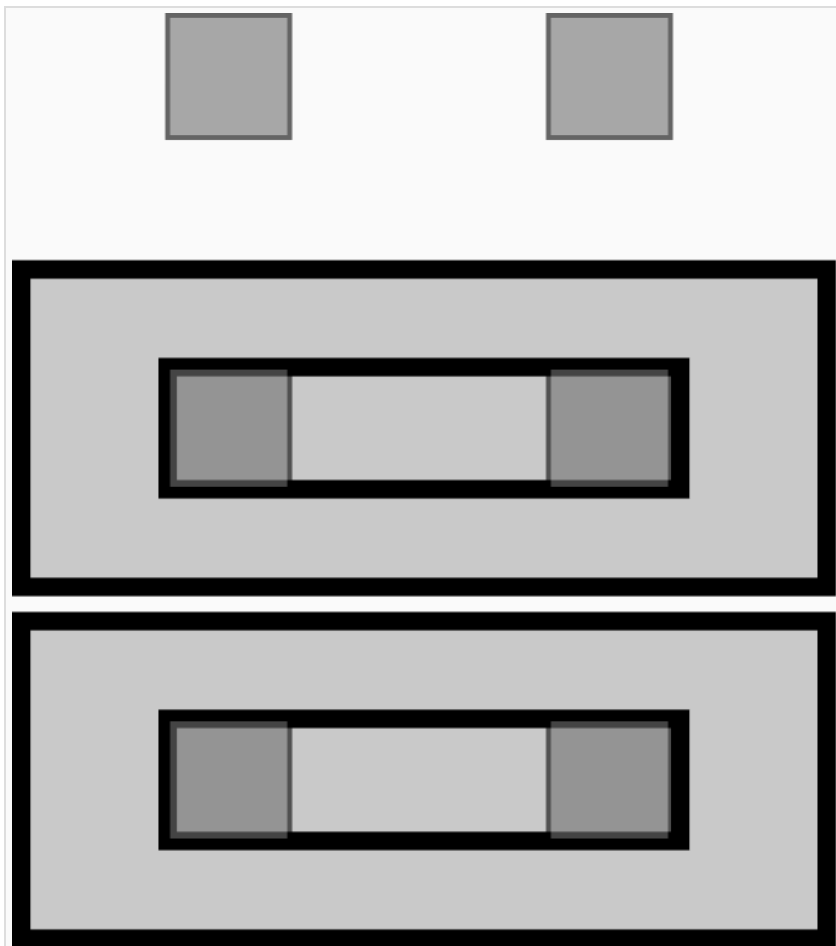


image28.png

Mack

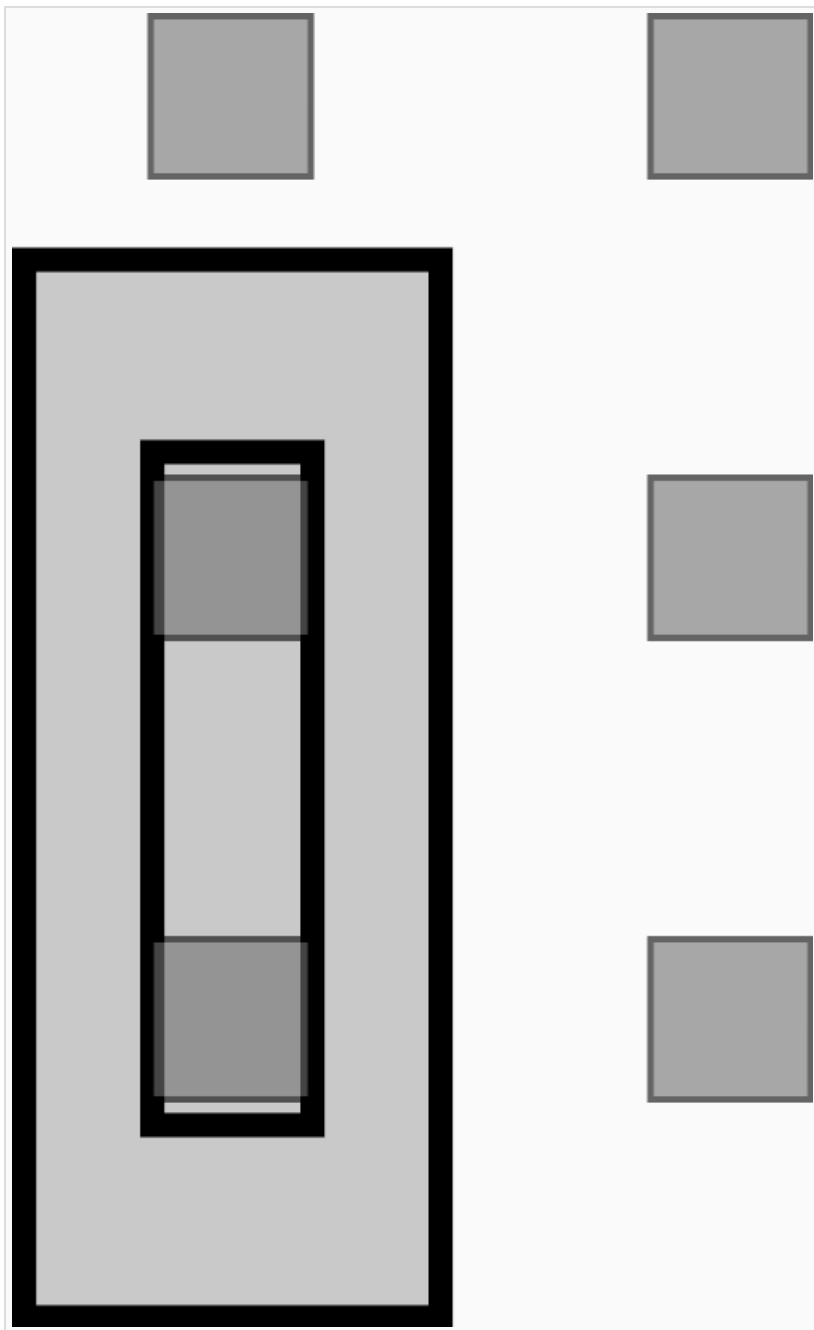


image30.png

Kenworth

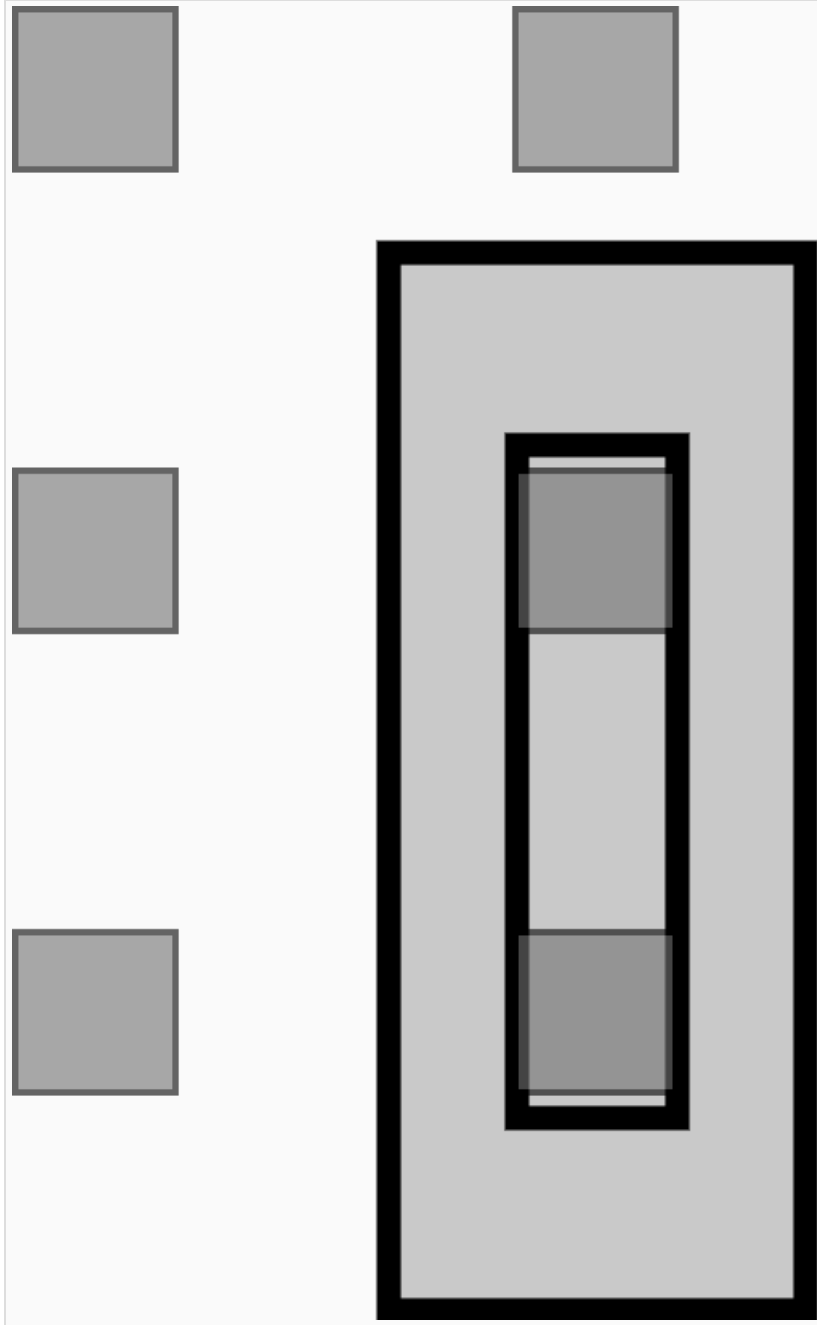


image32.png

Custom

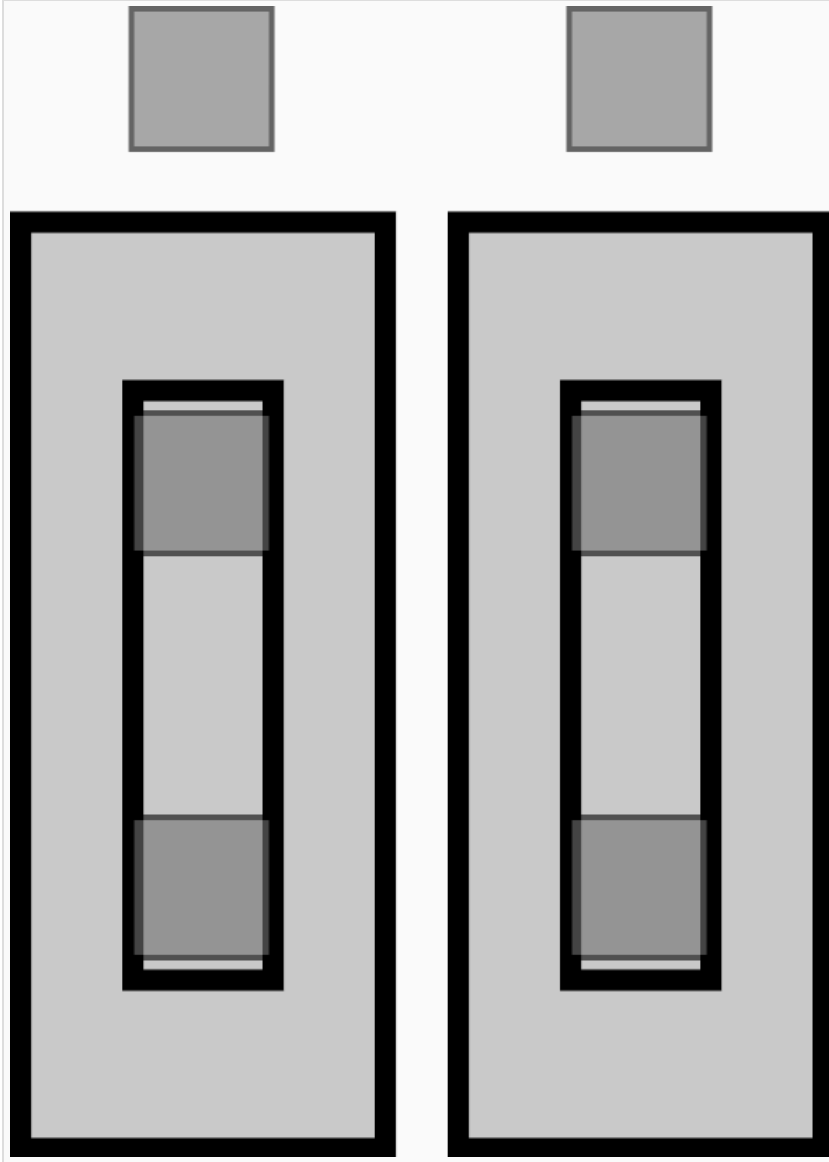


image34.png

Wiring Diagrams

Channel assignments may vary depending on your remote control system.

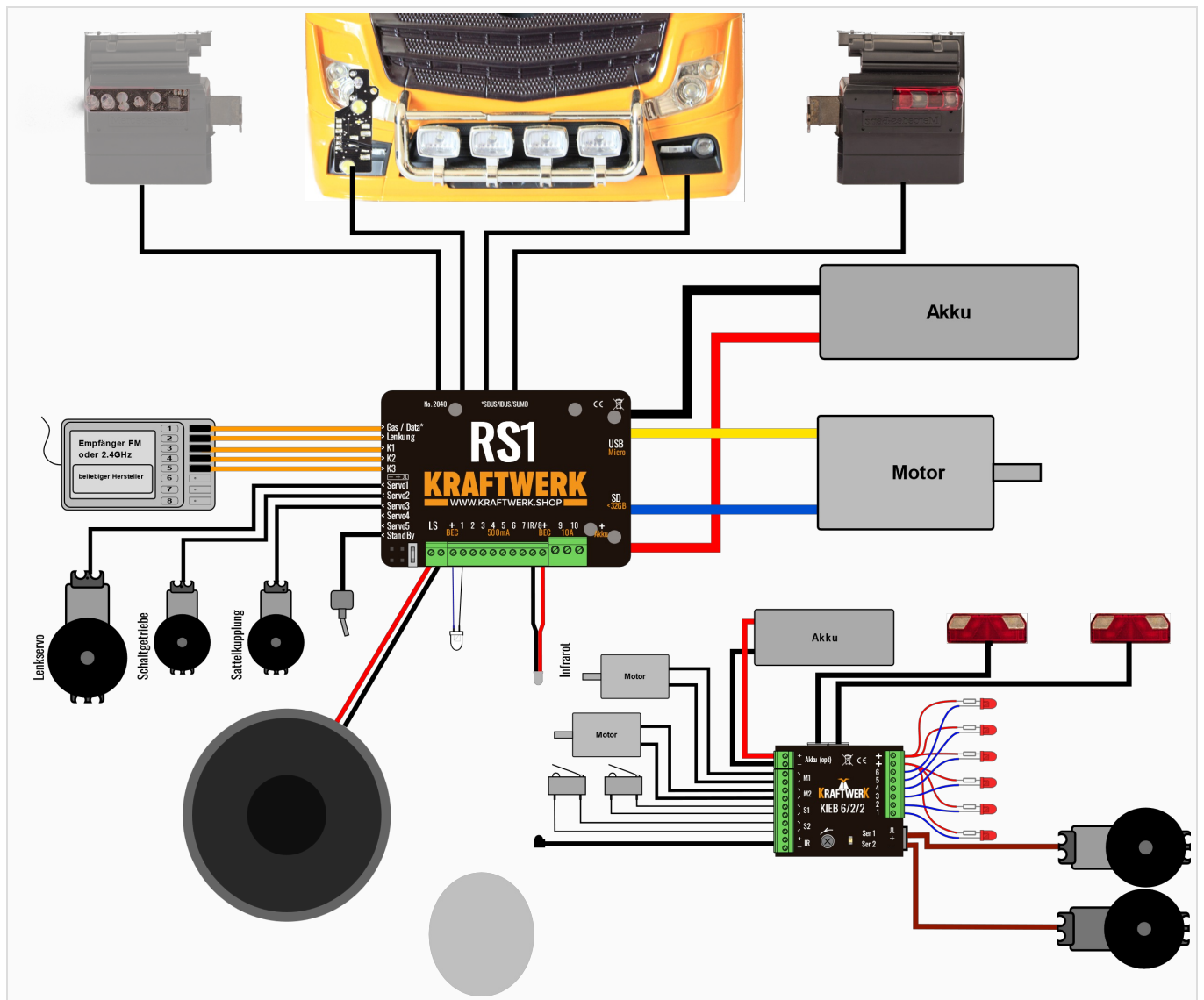


image36.png

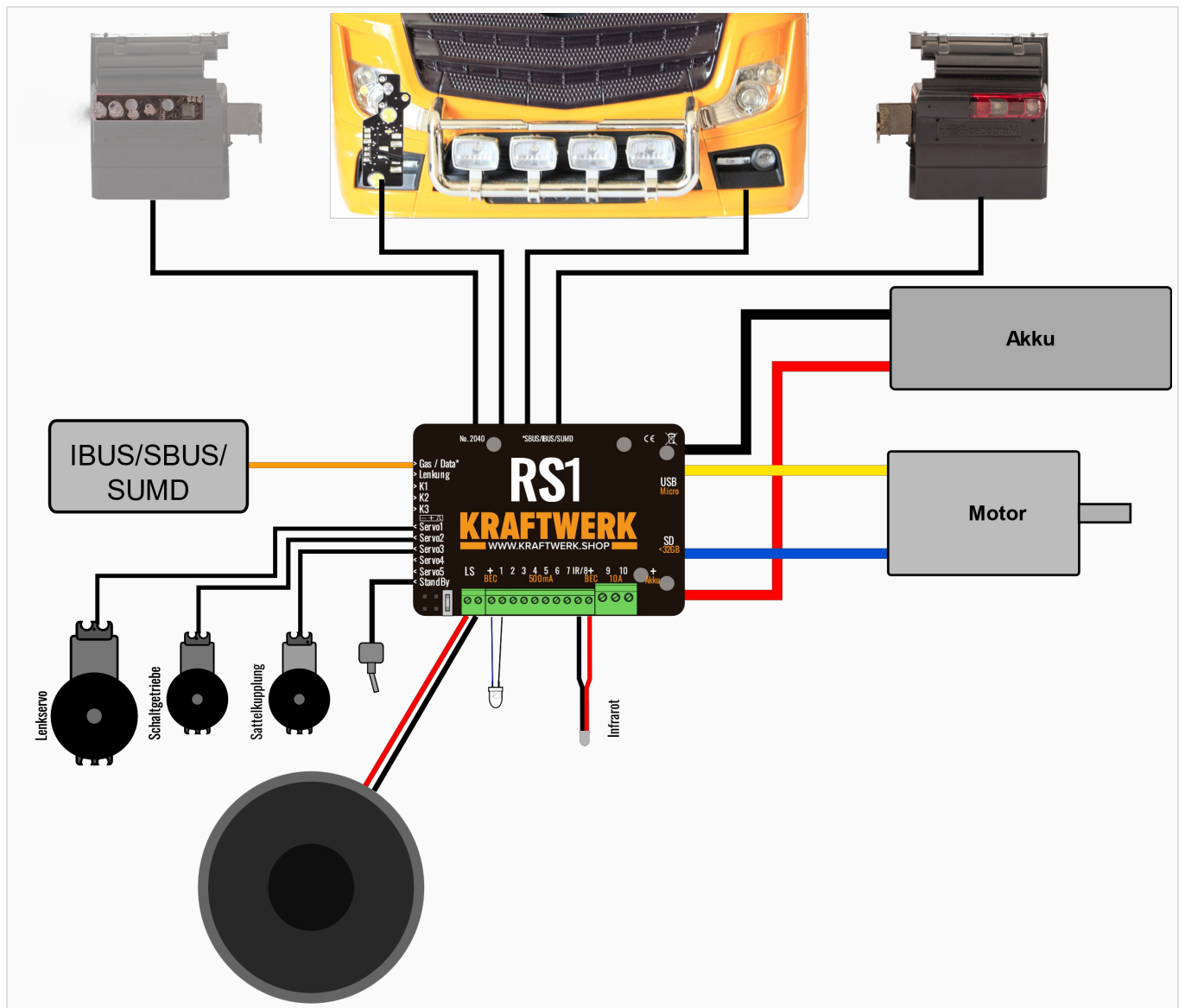


image38.png

ControlPanel Software



image40.png

Menu Symbols



image41.png

- **Back to main overview**
- **Reconnect:** Searches for new modules.
- **Reload module:** Discards all unsaved settings.
- **Open settings/project**
- **Save settings/project:** Overwrites current settings.
- **Save as:** Saves under a new filename.
- **Write all settings to module**

Model Functions

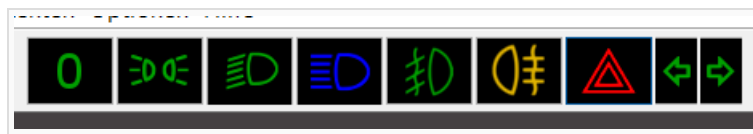


image42.png

- Motor control, battery voltage, and BEC current monitoring.
- Motor start/stop.
- Activate engine brake.
- Activate load sound.
- Battery voltage and BEC current display.

Assistant Symbols

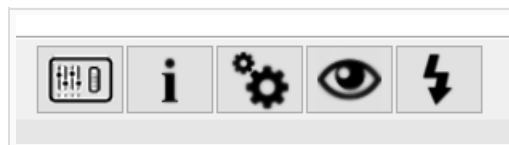


image44.png

- Open Live Data
- Open Device Information
- Open System Settings
- Show Output
- Open Live Update

Live Data

The Live Data view provides real-time visualization of module status, including stick positions and switch states. Channel assignments can be adjusted here if using IBUS, SBUS, or SUMD receivers. Individual channels can also be learned by clicking the "Learn" button and following the instructions.

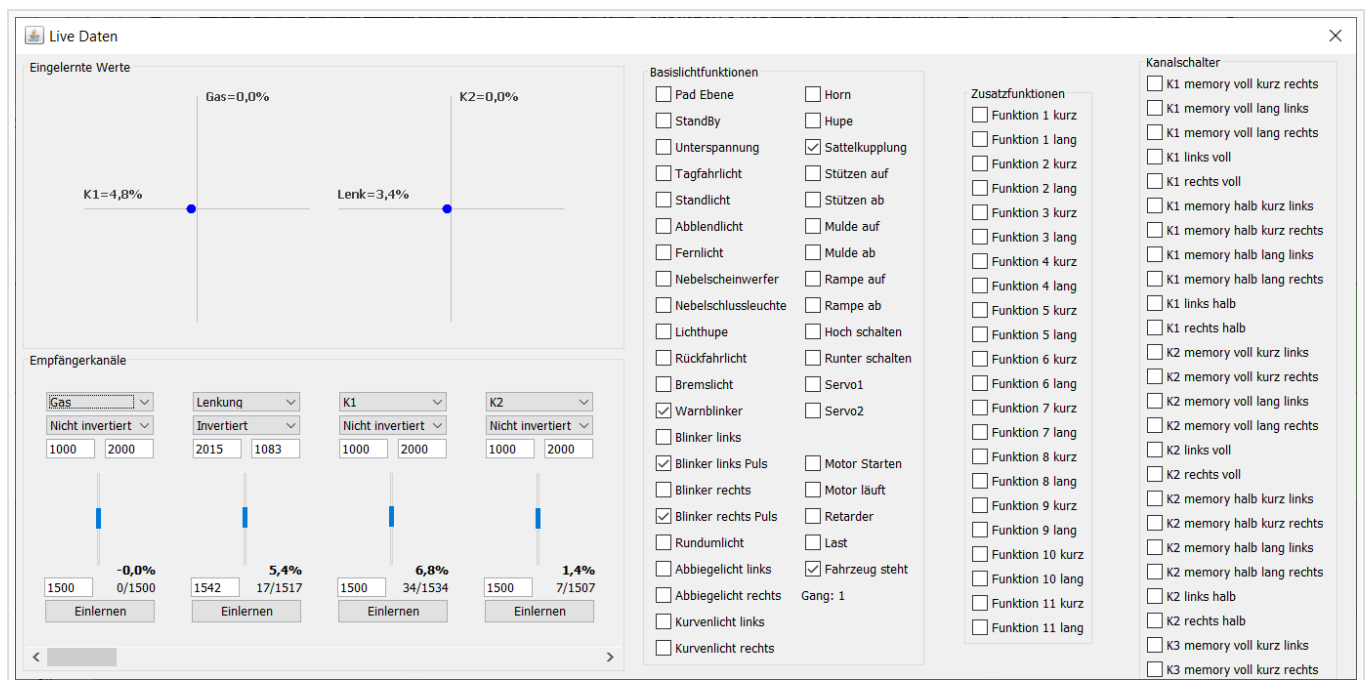


image45.png

The right side of the view may differ depending on whether a control pad or SUMD remote control is used.



image46.png

Device Information

Displays module-specific information such as:

- Name
- Software version
- Sound engine version
- Bootloader version
- Number of outputs
- Production date
- Device type

System Settings

Basic module settings that significantly influence model behavior can be configured here.

The screenshot shows the 'Systemeinstellungen' (System Settings) window for a device named 'RS1'. The window is organized into two columns of settings. At the top, there is a 'Gerät' (Device) dropdown menu set to 'RS1' and a save icon. The settings are as follows:

Setting	Value
Name	RS1
Abschaltspannung [V]	6,4 V
Kurze/Lange Betätigung: Lang ab	300 ms
Halbausschlag von 5% bis	20 %
Infrarot Sendeleistung	Aus
Motorstart automatisch	☒
Tagfahrlicht immer an	☐
Blinker Ein	250 ms
Fahrtenreglertyp	Bremse getrennt
Bremslicht Sensitivität	50 %
Rückfahrlicht Nachleuchten	500 ms
Blinker Schwellwert	75 %
Motor läuft sofort [aktiv]	☐
Pad Licht&Sound Fernlicht	28 %
Hydraulikdrehzahlerhöhung	20 %
Einlernmodus aktiv	☒
Modellnummer	1
Lautstärke	29 %
Vollausschlag ab [] bis 100%	80 %
Motorprofil	TAMIYA 540 Standard
Motorstop nach	10 s
Blinker Aus	250 ms
Rückwärtsfahren	---
Bremslicht Nachleuchten	30 %
Lenkungstyp	Normal
Kurvenlichttyp	Immer aktiv
Partymodus	Alle Rundumlicht/Warnblinker/Lang/Kurz/Lich...
Pad Licht&Sound Nebelscheinwerfer	-28 %
Hydrauliknachlauf	250 ms
Belegung K1	RS1 K1 Standard
Belegung K2	RS1 K2 Standard
Belegung K3	Pad Licht And Sound
Belegung K4	Keine
Belegung K5	Keine
Belegung K6	Keine
Belegung K7	Keine
Belegung K8	Keine
Belegung K9	Keine
Belegung K10	Keine

image47.png

- **Name:** Module name for identification, especially when using multiple KLB modules.
- **Learning Mode Active:** Enables/disables manual learning mode (throttle and steering sticks to endpoints).
- **Cutoff Voltage:** Default 6.4 V for 7.2 V NiMH/NiCd batteries or 2S LiPo. Set to 9.6 V for 3S LiPo.
- **Model Number:** Switch between up to 3 models.
- **Short/Long Press Threshold:** Defines duration for a press to count as long.
- **Volume:** Initial sound volume; adjusts automatically if controlled via potentiometer or stick.
- **Half Travel Threshold:** Defines the range considered as half travel.
- **Full Travel Threshold:** Defines the range considered as full travel.
- **Infrared Transmission Power:** Set 1 for close LED spacing, 3 for wider spacing.

- **Motor Profile:** Influences start behavior and cruise control.
- **Automatic Motor Start:** Starts motor when throttle stick is slightly moved forward.
- **Motor Stop Delay:** Time in seconds before motor stops after throttle inactivity.
- **Daytime Running Lights Always On:** If unchecked, daytime lights activate only when parking lights are on.
- **Blinker On/Off Durations:** Adjust pulse on/off times to control blink speed.
- **Speed Controller Type:** Currently supports only "Separate Brake".
- **Reverse Driving:** Can link warning and turn signals to reverse lights.
- **Brake Light Sensitivity:** Adjusts sensitivity of intelligent brake light.
- **Brake Light Afterglow:** Controls brake light duration after stopping.
- **Reverse Light Afterglow:** Controls reverse light duration after stopping.
- **Steering Type:** Automatic blinkers or manual operation.
- **Blinker Threshold:** Defines activation and reset points for automatic blinkers.
- **Cornering Light Type:** Always on or only with headlights.
- **Motor Runs Immediately:** Motor runs immediately after power on or requires manual start.
- **Party Mode:** Defines functions activated by the party mode button on the pad.
- **Pad Light & Sound Adjustments:** Fine-tuning of smallest travel ranges for left/right halves of the control pad.
- **Hydraulic RPM Increase:** Defines motor RPM increase when a hydraulic channel is active.
- **Hydraulic Afterrun:** Defines delay after hydraulic channel deactivation.

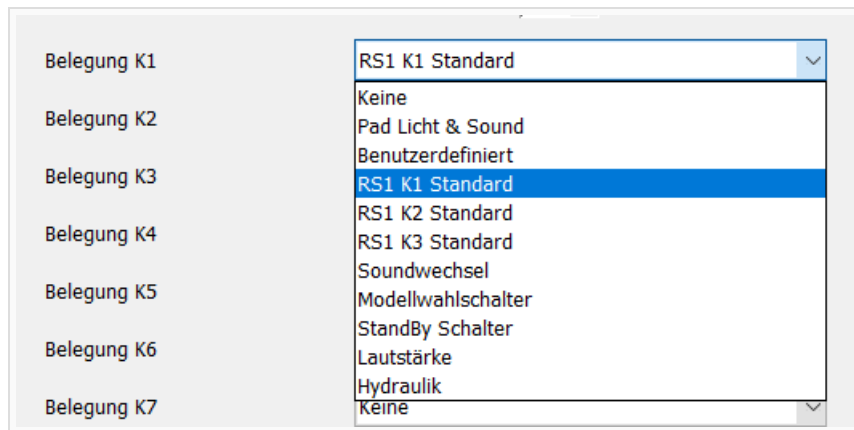


image48.png

Channel Assignments K1 – K32

- **None:** Channel not used.
- **Pad Light & Sound:** Used when control pad is connected to this channel.
- **Custom:** User-defined assignments for half/full, short/long travel as memory or direct function.

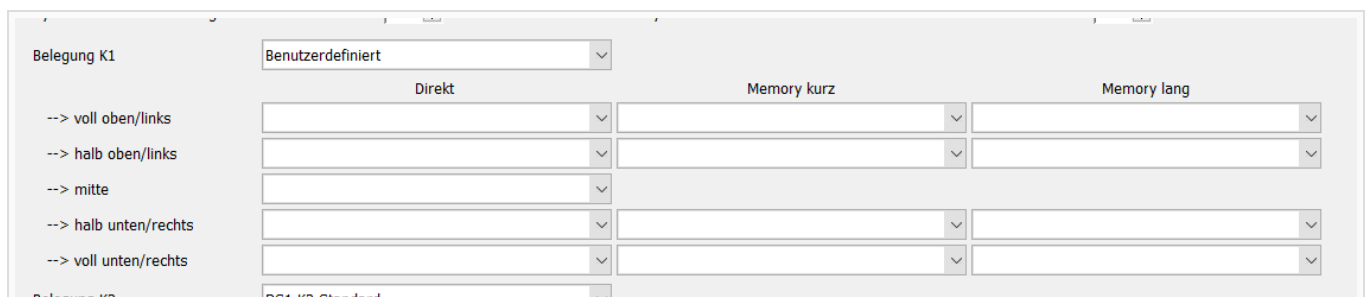


image49.png

Possible Settings

Belegung K1

Benutzerdefiniert

Direkt

Memory kurz

Memory lang

--> voll oben/links

--> halb oben/links

--> mitte

--> halb unten/rechts

--> voll unten/rechts

Belegung K2

Belegung K3

Belegung K4

Belegung K5

Belegung K6

Belegung K7

Belegung K8

Belegung K9

Belegung K10

Belegung K11

Licht Aus

Standlicht

Abblendlicht

Fernlicht

Stand -> Abblendlicht

Stand -> Abblend -> Fernlicht

Lichthupe

Nebelscheinwerfer

Nebelschlussleuchte

Nebelscheinwerfer -> Nebelschlussleuchte

Warnblinker

Blinker rechts

Blinker links

Sattelkupplung

Rundumlicht

Horn

Hupe

Stützen auf

Stützen ab

Mulde auf

Mulde ab

Rampe auf

Rampe ab

Motor läuft

Keine

image50.png

- RS1 K1 Standard, RS1 K2 Standard, RS1 K3 Standard: Default values for RS1 channels K1-K3.
- Sound Change: Switch between sounds via button press.
- Model Selector: Switch between models; current model must be defined in system settings.
- Standby Switch: Standby mode similar to the physical RS1 switch.
- Volume: Direct volume control via remote control.
- Hydraulics: Triggers motor sound RPM increase.

RS1 Sound Settings

Kraftwerk - ControlPanel 2023.2 - RS1 Soundeinstellungen

Datei Module Beispiele System Programmierhilfen Assistenten Optionen Hilfe

Projektname

Min Drehzahl

Max Drehzahl

Dynamik

Max Beschleunigung

Max Auslaufen

Max Beschleunigung

Max Auslaufen

Motor Start

Last

Motorbremse

U/min

Turbo

Beschleunigung

Einstellungen speichern

Sounds speichern

Sync SD

Standardtyp

Umschaltgeschwindigkeit

Last %

Motorbremse %

Motor Sound Standard

100

100

ACHTUNG: Die Tonqualität am PC ist bei mitgelieferten Sounds reduziert!

Fanfare

Play

Vol

100%

/sdcard/Scania/fanfare0.snd

Öffnen

/sdcard/Scania/fanfare0.snd

00:00:872

Speichern

Löschen

Horn

Play

Vol

100%

/sdcard/Scania/ahorn.snd

Öffnen

/sdcard/Scania/ahorn.snd

00:03:606

Speichern

Löschen

Luftbremse Aus

Play

Vol

10%

/sdcard/Scania/airbrakeoff0.sr

Öffnen

/sdcard/Scania/airbrakeoff0.snd

00:00:700

Speichern

Löschen

Luftbremse Aus

Play

Vol

10%

/sdcard/Scania/airbrakeoff1.sr

Öffnen

/sdcard/Scania/airbrakeoff1.snd

00:00:687

Speichern

Löschen

Motor Sound Standard

Play

Vol

150%

/sdcard/Scania/e0.snd

Öffnen

/sdcard/Scania/e0.snd

00:01:931

Speichern

Löschen

☒ Motor Standard

☒ Motor Last

☐ Motor Verzögerung

☐ Motor Motorbremse

min RPM

400

ref RPM

500

max RPM

600

image51.png

Basic Settings

Projektname

Drehzahlbereich

Motorcharakteristik

Turbocharakteristik

Min Drehzahl

Max Drehzahl

Dynamik

Max Beschleunigung

Max Auslaufen

Max Beschleunigung

Max Auslaufen

/sdcard/Scania/scania.ecfg

500

2.400

10

40

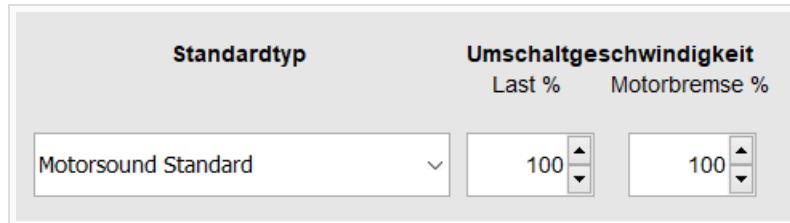
10

35

2

image52.png

- **Project Name:** Select projects, including preloaded sounds.
- **RPM Range Min/Max:** Defines motor sound RPM range.
- **Dynamics:** Defines how dynamically the sound reacts.
- **Motor Characteristics:** Sets max acceleration and deceleration speeds.
- **Turbo Characteristics:** Similar to motor characteristics but specific for turbo sounds.
- **Engine Brake Transition Speed %:** Speed of sound transition between normal and load states.



Standardtyp: Motorsound Standard

Umschaltgeschwindigkeit: Last % 100, Motorbremse % 100

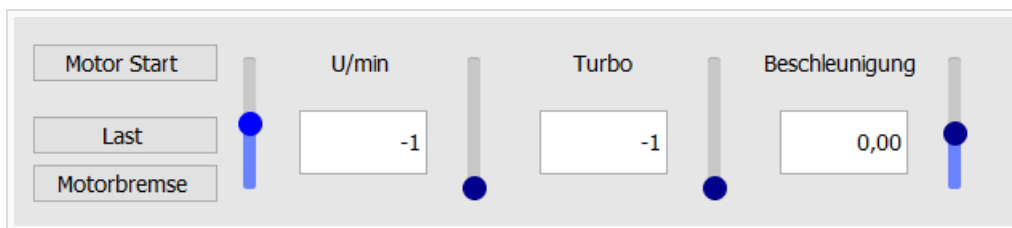
image53.png

- **Default Sound Type:** Preset sound type when adding new sounds.
- **Load Transition Speed %:** Speed of sound transition between normal and engine brake states.

ACHTUNG: Die Tonqualität am PC ist bei mitgelieferten Sounds reduziert!

image54.png

Note: Sounds on the PC are compressed for performance and copyright reasons. The module plays them in full quality.



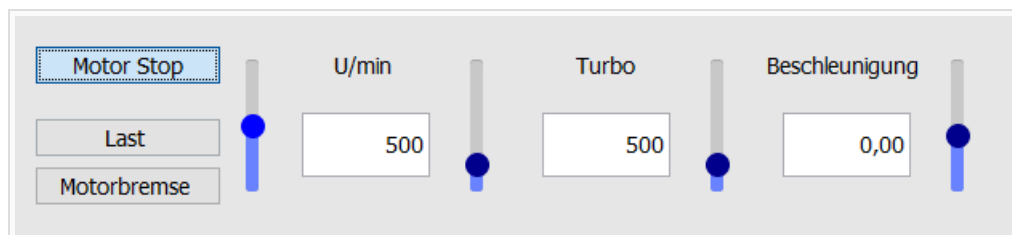
Motor Start, Last, Motorbremse

U/min: -1, Turbo: -1, Beschleunigung: 0,00

image55.png

Sound simulation on PC:

Click "Motor Start" to play the start sound and begin motor sound playback.

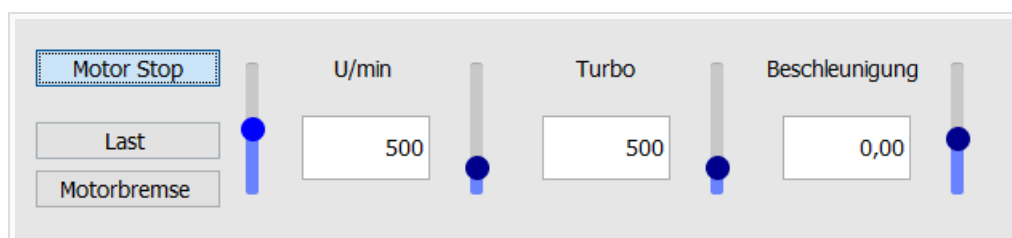


Motor Stop, Last, Motorbremse

U/min: 500, Turbo: 500, Beschleunigung: 0,00

image56.png

Moving the left slider up increases RPM; moving it down decreases RPM or engages engine brake.



Motor Stop, Last, Motorbremse

U/min: 500, Turbo: 500, Beschleunigung: 0,00

image56.png

Bars two to four visualize motor and turbo RPM and acceleration but cannot be adjusted directly. Active sounds are marked with playback position and volume based on RPM.

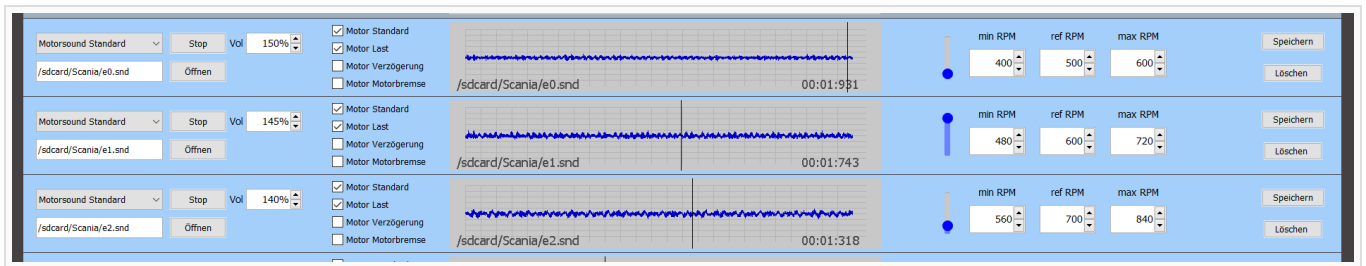


image57.png

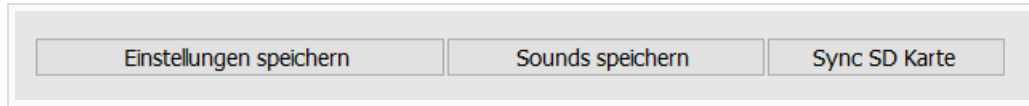


image58.png

Saving Settings

- Save Basic Settings
- Save Sounds
- Sync SD Card: Resets all sound settings.

Sound Types

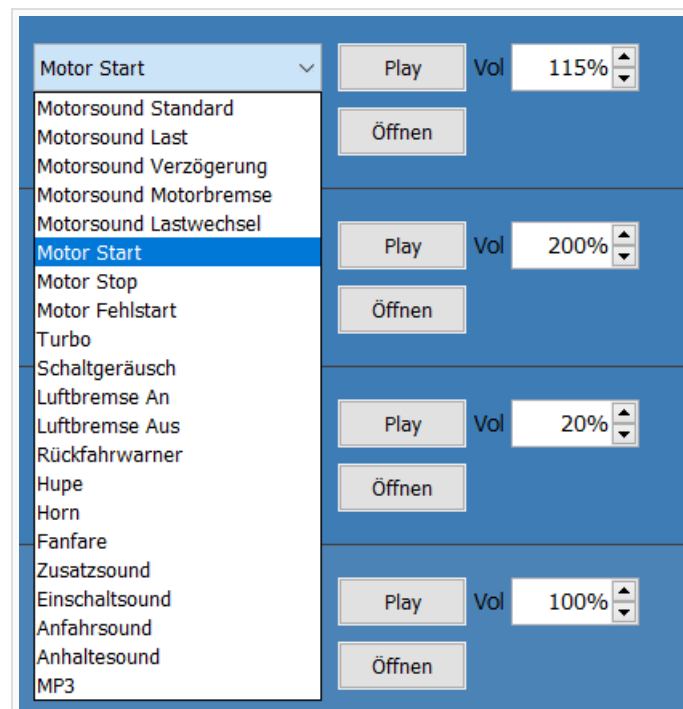


image59.png

- **Motor Sound Standard:** Acceleration and continuous driving.
- **Motor Sound Load:** Acceleration and driving under load.
- **Motor Sound Deceleration:** Vehicle slowing down.
- **Motor Sound Engine Brake:** Vehicle slowing with engine brake.
- **Motor Load Change:** Played once when driving state changes.
- **Motor Start/Stop:** Played when motor starts or stops; can be load-dependent.
- **Motor False Start:** Motor fails to start sound.
- **Turbo:** Turbo whistle sound.
- **Gear Shift Sound:** Played during gear changes.
- **Air Brake On/Off:** Sounds for air brake closing and opening.
- **Reverse Warning:** Played when reverse gear is engaged.
- **Horn/Horn/Fanfare:** Played via horn button, switchable.

- **Additional Sounds:** Custom sounds triggered by auxiliary functions.
- **Power-On Sound:** Played after module startup.
- **Start Sound:** Played when vehicle starts moving; usually unnecessary if enough standard motor sounds are present.
- **Stop Sound:** Played when vehicle stops.
- **MP3:** MP3 files can be defined; only one MP3 plays at a time.

Motor Sound Settings

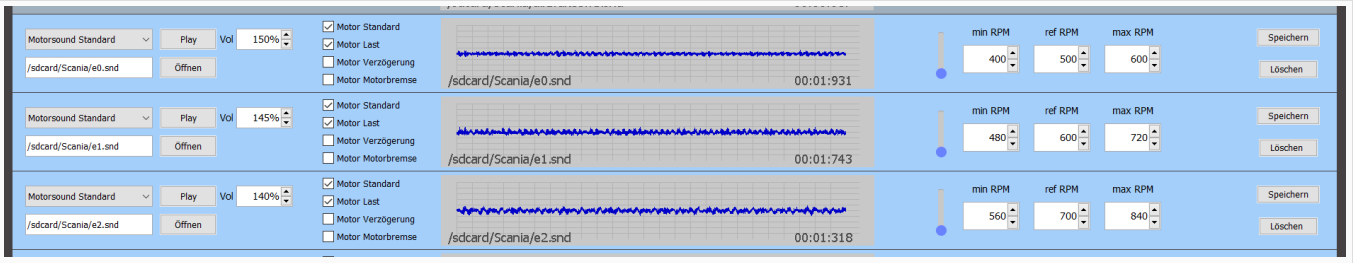


image60.png

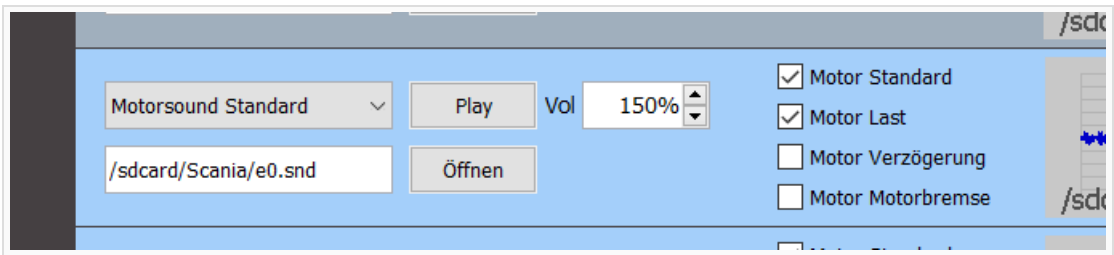


image61.png

- **Motor Sound Presets:** Standard, Load, Deceleration, Engine Brake; checkboxes define active states.
- **Filename on SD Card:** Kraftwerk internal format uses .snd extension.
- **Play:** Play the sound.
- **Open:** Load sound files.
- **Volume:** Base sound volume, adjusted dynamically during playback.

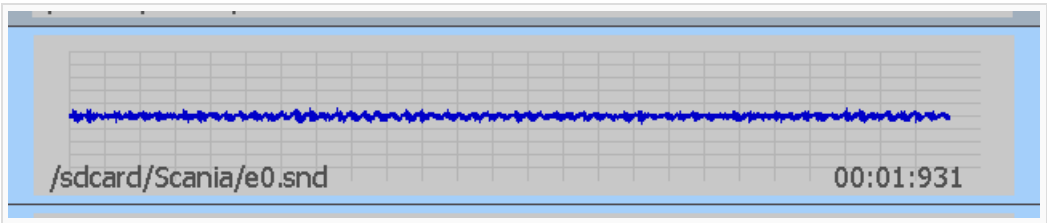


image62.png

Sound visualization includes filename and snippet length.

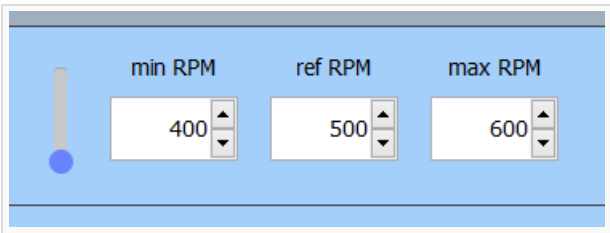


image63.png

- **Min RPM:** RPM at which sound fades in.
- **Reference RPM:** RPM at which sound plays at full volume without pitch shift.
- **Max RPM:** RPM at which sound fades out.

Each sound snippet fades in/out and pitch-shifts according to the RPM range.

Additional Sounds

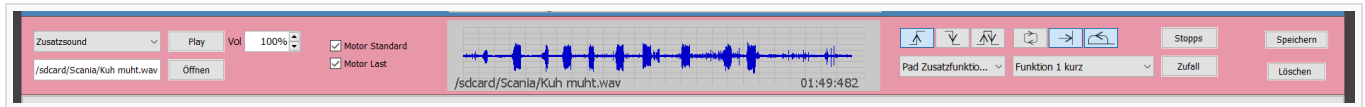


image64.png

The left half mirrors motor sound settings: sound type, filename, play, open, volume, and checkboxes. Only two checkboxes are available here: Motor Standard and Motor Load, defining playback with or without load.

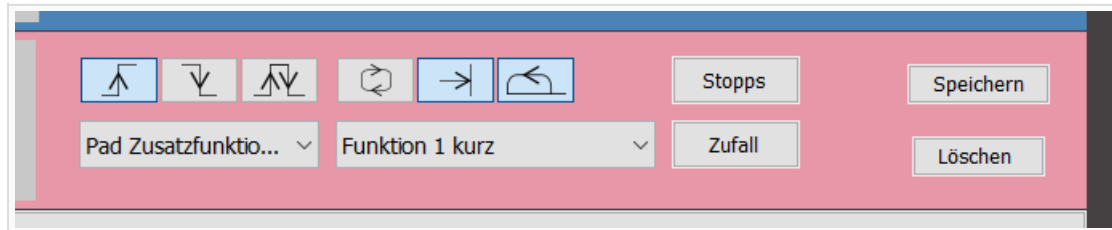


image65.png

Trigger Settings

- Trigger Types: Rising edge, falling edge, both edges.
- Playback Types: Loop, always play, return to start or continue if stopped.
- All combinations are possible.

Available triggers depend on whether a control pad or SUMD remote is used. Functions correspond to those in Live Data.

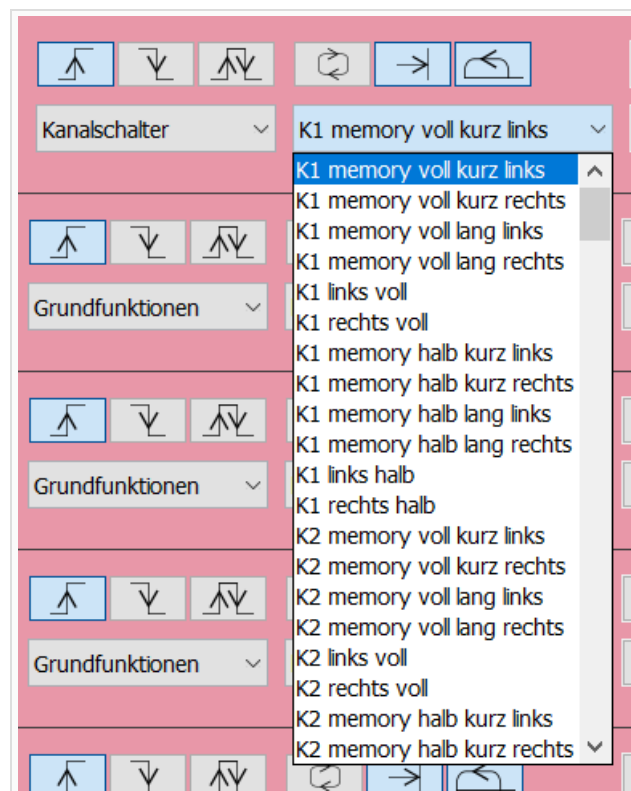


image66.png

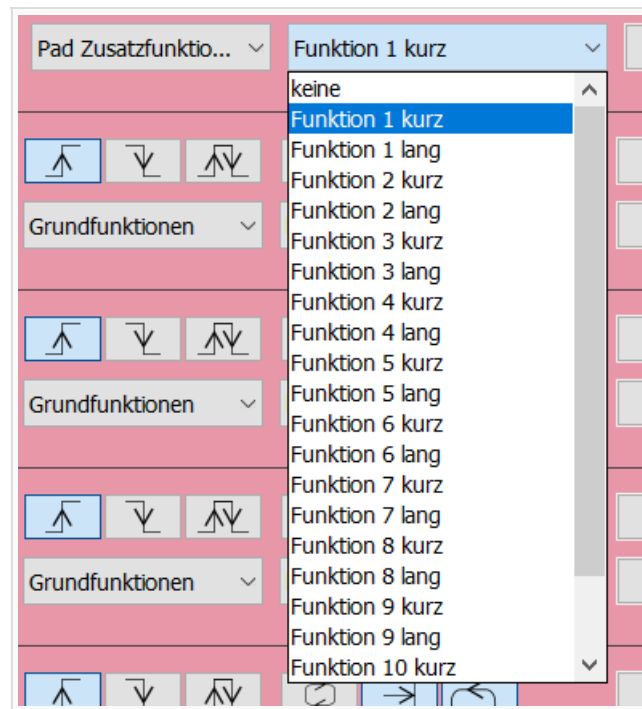


image67.png

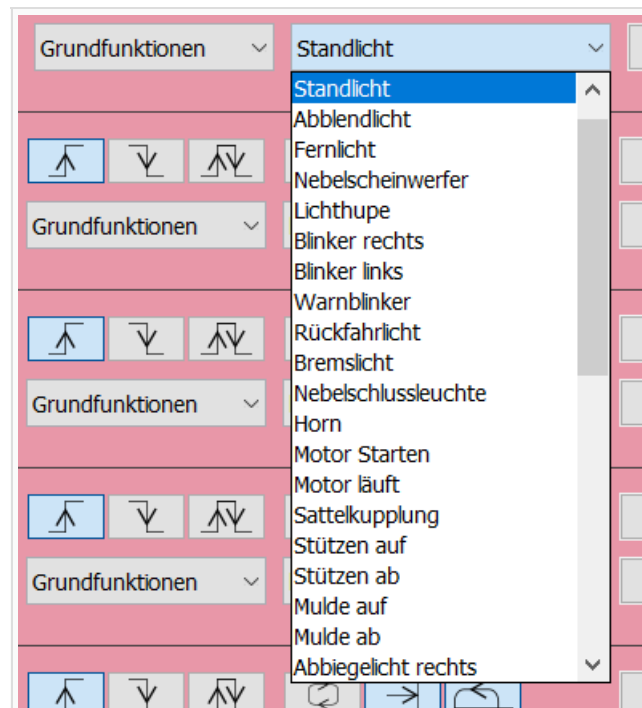


image68.png

Stop Points

Stop points can be defined in seconds for each sound. If "always play" is active, the sound plays until the next stop point or continues if the function remains active.

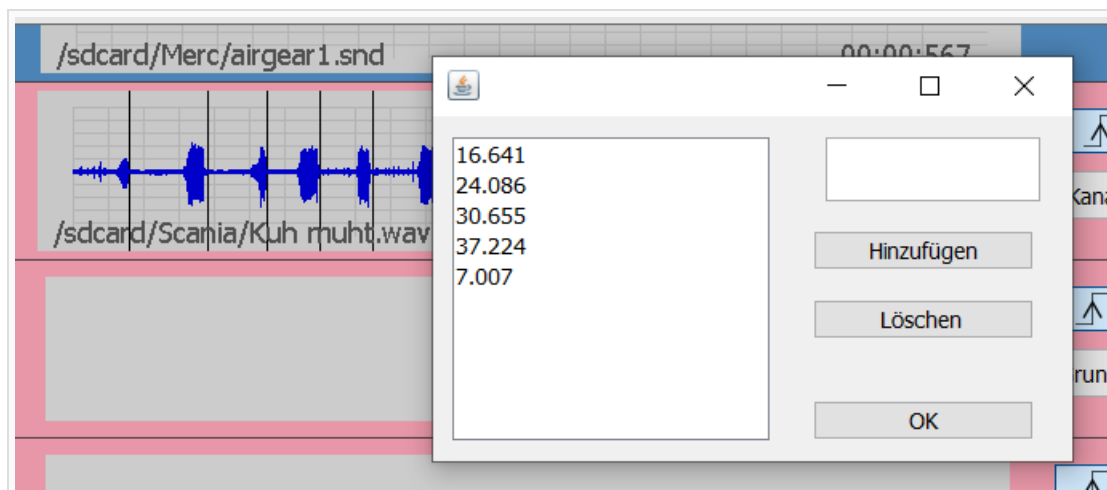


image69.png

Random Playback

Sounds can be set to start randomly every x to y seconds while active.

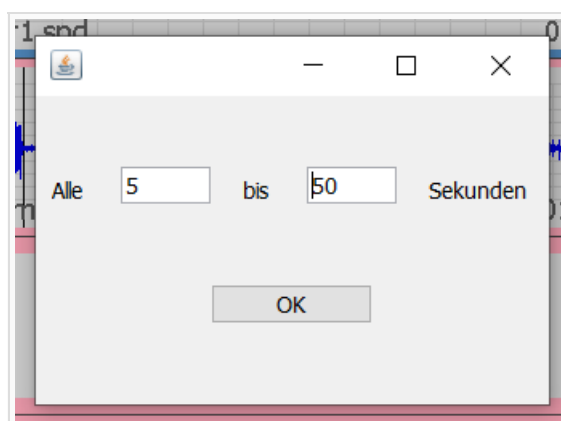


image70.png

Adding New Sounds

Supported sound types include all common audio and video formats. The ControlPanel automatically converts files to the required format and bitrate.

Examples: WAV, MP3, OGG, MP4, SND.

To add a new sound, click the "New Sound Entry" button at the bottom:

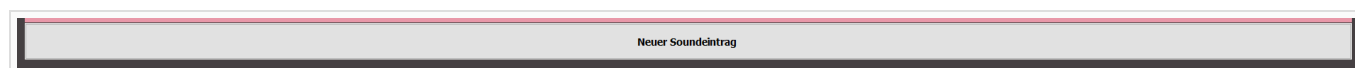


image71.png

A new row is created with the default sound type from basic settings. Use the "Open" button to load files.

Alternatively, drag and drop files into the ControlPanel:

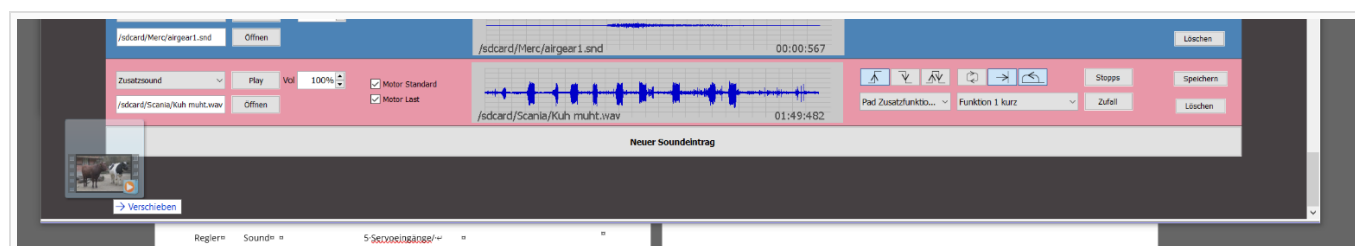


image72.png

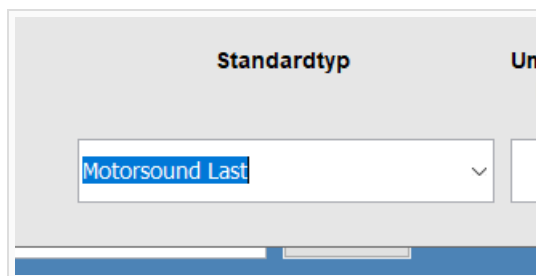


image73.png

Dragging a file into the sound visualization area replaces the existing sound instead of adding a new one.

For motor sounds, you can preselect the default type (e.g., Motor Load) and drag multiple files simultaneously. If filenames contain a 3-5 digit number, it will be used as the reference RPM, and the RPM ranges will be set accordingly. These can be fine-tuned later in ControlPanel before uploading to the module.

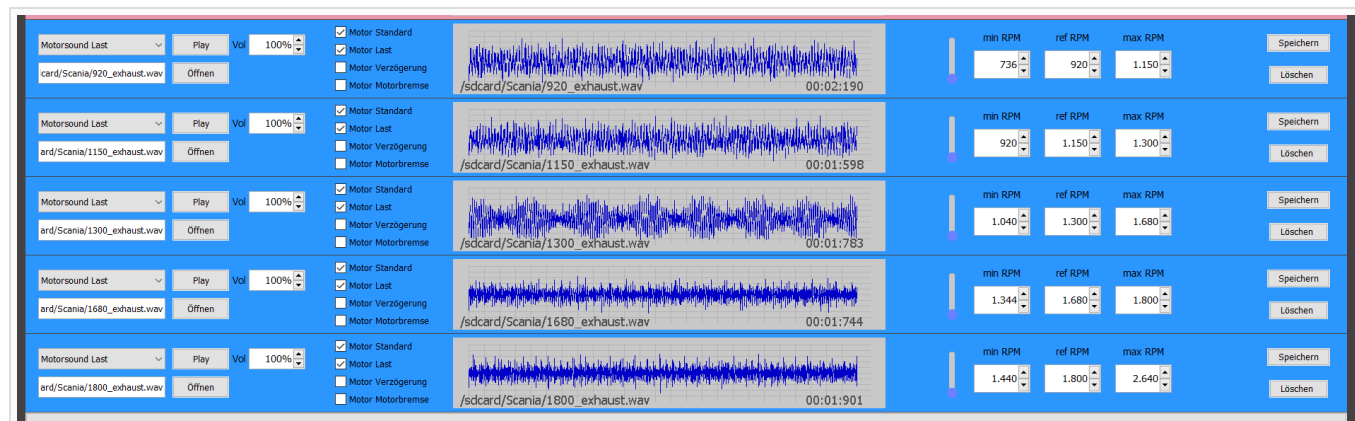


image74.png

Appendix: Naming Conventions

Brand	Type	Master/Bus Node	Inputs/Outputs	Additional Info
Kraftwerk	Controller		5 servo inputs / 5 servo outputs	
Kraftwerk	Light	Master	4 servo inputs / 16 switching outputs	
Kraftwerk	Light	Master	4 servo inputs / 12 switching outputs	500 mA per output
Kraftwerk	Light	Bus Node		500 mA per output
Kraftwerk	Servo	Bus Node	2 outputs	
Kraftwerk	Motor	Bus Node	1 output	10 Ampere

Glossary

- **ControlPanel:** Free PC software to configure the system.
- **Control Pad:** Mounted on the remote control, enabling convenient system operation.
- **Parameter:** Adjustable value or setting.